TMUA / PRACTICE PAPER

# Set A Paper 2

Mathematical Reasoning 20 questions · 75 minutes

## Questions


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*Proof & Error Analysis*

**1.** Let $x$ and $y$ be real numbers.

A student attempts to prove the following claim:

$$\textbf{Claim: If } x + y \geq 2 \text{ then } x^2 + y^2 \geq 2$$

The student's attempt is as follows:

**Line 1:** Since $x + y \geq 2$, at least one of $x$ and $y$ is at least 1.
Assume $x \geq 1$; the other case is the same with $x$ and $y$ swapped.

**Line 2:** From $x + y \geq 2$ we get $y \geq 2 - x$

**Line 3:** Therefore $y^2 \geq (2 - x)^2$

**Line 4:** Hence $x^2 + y^2 \geq x^2 + (2 - x)^2$

**Line 5:** Now $x^2 + (2 - x)^2 = 2(x - 1)^2 + 2$, and this is at least 2.

**Line 6:** Therefore $x^2 + y^2 \geq 2$, as claimed.

Which of the following best describes the argument?

**A** Line 1 contains the first error.

**B** Line 2 contains the first error.

**C** Line 3 contains the first error.

**D** Line 4 contains the first error.

**E** Line 5 contains the first error.

**F** The argument is a correct proof of the claim.

**G** The argument contains no error, but the claim is false.

*Trigonometry*

**2.** In quadrilateral $ABCD$, the diagonal $AC$ divides the figure into two right-angled triangles, $\triangle ABC$ and $\triangle ACD$.

- In $\triangle ABC$, $\angle ABC = 90°$, $\angle BAC = 30°$, and $AB = \sqrt{6}$ cm
- In $\triangle ACD$, $\angle ACD = 90°$, and $\tan(\angle CAD) = \dfrac{1}{\sqrt{2}}$

What is the area of quadrilateral $ABCD$, in $\text{cm}^2$?

**A** $\sqrt{3} + \sqrt{2}$

**B** $\sqrt{3} + 2\sqrt{2}$

**C** $2\sqrt{3} + \sqrt{2}$

**D** $2\sqrt{3} + 2\sqrt{2}$

**E** $\sqrt{3} + 4\sqrt{2}$

**F** $2\sqrt{3} + 4\sqrt{2}$

*Proof & Error Analysis*

**3.** Consider the following claim and attempted proof.

**Claim:** Let $f(x)$ be a quadratic polynomial. If $\displaystyle\int_{-1}^{1} f(x)\,\mathrm{d}x = 0$, then the graph of $y = f(x)$ must intersect the $x$-axis at two distinct points in the interval $-1 < x < 1$.

Attempted proof:

**Line 1** Let $f(x) = ax^2 + bx + c$, where $a, b, c$ are real constants with $a \neq 0$

**Line 2** Evaluating the integral gives $\displaystyle\int_{-1}^{1} f(x)\,\mathrm{d}x = \frac{2a}{3} + 2c = 0$, so $c = -\frac{a}{3}$

**Line 3** The discriminant is $b^2 - 4ac = b^2 + \frac{4a^2}{3} > 0$, so $f(x) = 0$ has two distinct real roots

**Line 4** Writing these roots as $x_1$ and $x_2$, we have $x_1 x_2 = \frac{c}{a} = -\frac{1}{3}$

**Line 5** Since $|x_1 x_2| = \frac{1}{3} < 1$, it follows that $|x_1| < 1$ and $|x_2| < 1$

**Line 6** Therefore, both roots lie in the interval $-1 < x < 1$

Which line of the proof contains the first error?

**A** Line 1

**B** Line 2

**C** Line 3

**D** Line 4

**E** Line 5

**F** Line 6

**G** There is no error, the proof is correct as written.

*Exponentials & Logarithms*

**4.** Let $a$ be a positive real number with $a \neq 1$

The equation

$$\log_a(3x-1) = 2 + \log_a(x+1)$$

has exactly one real solution $x$

What is the complete range of all the possible values of $a$?

**A** $0 < a < \sqrt{3}$ and $a \neq 1$

**B** $a > 1$

**C** $a > 0$ and $a \neq 1$

**D** No such $a$ exists

**E** $a = 3$

**F** $a = \frac{1}{3}$

**G** $a > 0$, $a \neq 1$ and $a \neq \sqrt{3}$

*Number*

**5.** A positive integer $n$ can be written as $n = 2^a 3^b$ where $a$ and $b$ are non-negative integers.
The number $n$ has exactly 12 positive factors, counting 1 and $n$ itself.
Given that $n < 200$, what is the greatest possible value of $n$?

**A** 72

**B** 96

**C** 108

**D** 144

**E** 162

**F** 486

*Proof & Error Analysis*

**6.** Let $a$ and $b$ be real numbers. A student attempts to prove the following claim:

**Claim:** If $a+b$ and $ab$ are both integers, then $a$ and $b$ are both integers.

The student's attempt is as follows:

**I** $a$ and $b$ are the two solutions of $t^2-(a+b)t+ab=0$

**II** Write $s=a+b$ and $p=ab$, so $s$ and $p$ are integers and the two solutions are $t=\dfrac{s\pm\sqrt{s^2-4p}}{2}$

**III** Since $a$ and $b$ are real, $s^2-4p\geq 0$

**IV** Also $s^2-4p$ is an integer, so $\sqrt{s^2-4p}$ is either an integer or an irrational number.

**V** If $\sqrt{s^2-4p}$ were irrational, then $a$ would be irrational, and so $a+b$ would be irrational, contradicting the fact that $a+b$ is an integer.

**VI** Hence $\sqrt{s^2-4p}$ is an integer, so $s\pm\sqrt{s^2-4p}$ is an integer, and therefore $a$ and $b$ are both integers.

Which of the following best describes this attempt?

**A** It is completely correct.

**B** It is incorrect, and the first error is on line I.

**C** It is incorrect, and the first error is on line II.

**D** It is incorrect, and the first error is on line III.

**E** It is incorrect, and the first error is on line IV.

**F** It is incorrect, and the first error is on line V.

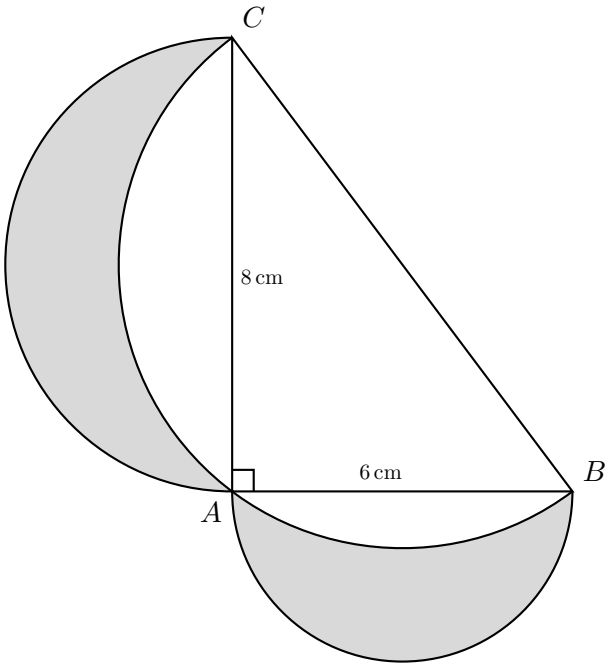
**G** It is incorrect, and the first error is on line VI.

*Geometry*

**7.** Triangle $ABC$ is right-angled at $A$, with $AB = 6\,\text{cm}$ and $AC = 8\,\text{cm}$.

Semicircles are drawn with $AB$, $AC$ and $BC$ as diameters, as shown. The semicircle on $BC$ is on the same side of the line $BC$ as $A$. The portions of the other two semicircles that lie outside the semicircle on $BC$ are shaded.

What is the total shaded area, in $\text{cm}^2$?




**A** 12

**B** 24

**C** 30

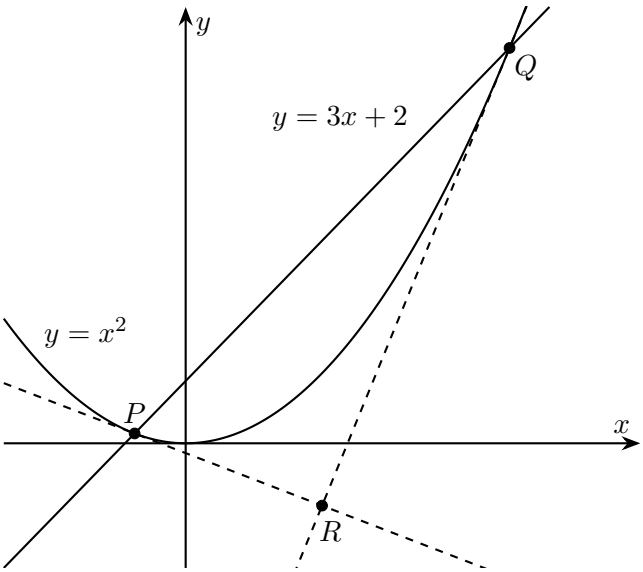
**D** 40

**E** 48

**F** 50

*Coordinate Geometry*

**8.** The line $y = 3x + 2$ meets the parabola $y = x^2$ at the points $P$ and $Q$.
The tangents to the parabola at $P$ and $Q$ meet at $R$, as shown.
What are the coordinates of $R$?




**A** $\left(\frac{3}{2}, -4\right)$

**B** $\left(\frac{3}{2}, -2\right)$

**C** $\left(\frac{3}{2}, -1\right)$

**D** $(3, -2)$

**E** $(1, -3)$

**F** $\left(2, -\frac{3}{2}\right)$

*Logic*

**9.** Consider the statement $(*)$:

$$\textbf{If } 0 \leq f(x) \leq 1 \text{ for all real } x \text{ with } 0 \leq x \leq 1, \textbf{ then}$$
$$\int_0^1 f(x)\,\mathrm{d}x \leq \int_0^1 (f(x))^2\,\mathrm{d}x \qquad (*)$$

Which of the following functions is a **counterexample** to $(*)$?

**A** $f(x) = 2x$

**B** $f(x) = x^2 + \frac{1}{2}$

**C** $f(x) = 1$

**D** $f(x) = 1 + x$

**E** $f(x) = 0$

**F** $f(x) = 5x(1 - x)$

**G** $f(x) = \sqrt{x} + \frac{1}{2}$

**H** $f(x) = \frac{1}{2}$

*Logic*

**10.** Consider the following three conditions on a positive real number $x$

(1) $x^2 - 3x + 2 < 0$
(2) $\log_{10} x > 0$
(3) $2^x > 4$

Which of the following statements is/are true?

**I:** Condition (2) holds only if at least one of conditions (1) and (3) holds

**II:** If condition (2) does not hold, then condition (3) does not hold

**III:** For condition (1) to hold, it is necessary but not sufficient that condition (3) does not hold

**A** None of them

**B** I only

**C** II only

**D** III only

**E** I and II only

**F** I and III only

**G** II and III only

**H** I, II and III

*Probability & Counting*

**11.** A three-digit code is formed by selecting three different digits from $1, 2, 3, 4, 5, 6$ and placing them in random order.

Each possible code is equally likely.

It is revealed that when the three digits in the code are multiplied together, the result is divisible by 6.

What is the probability that the code contains the digit 6?

**A** $\frac{1}{3}$

**B** $\frac{1}{2}$

**C** $\frac{3}{5}$

**D** $\frac{2}{3}$

**E** $\frac{3}{4}$

**F** $\frac{5}{6}$

*Proof & Error Analysis*

**12.** Consider the claim about cubic polynomials:

**Claim:** If a cubic polynomial $f$ has distinct roots $a$, $b$ and $c$, where $a < b < c$, then there exists $x$ with $a < x < b$ such that $f''(x) = 0$

The student attempts to prove the claim and produces the following argument:

**I** Since $f(a) = f(b) = 0,$ there is a number $p$ such that $a < p < b$ and $f'(p) = 0$

**II** Similarly, there is a number $q$ such that $b < q < c$ and $f'(q) = 0$

**III** The polynomial $f'$ is quadratic with roots $p$ and $q,$ so $f''\left(\frac{p+q}{2}\right) = 0$

**IV** Since $a < p < b < q < c,$ we have $p - b < 0 < q - b,$ and hence
$$\frac{p+q}{2} - b = \frac{(p-b)+(q-b)}{2} < 0$$

**V** Therefore the claim is true, the claimed $x$ can be chosen as $x = \frac{p+q}{2}$

Which line contains the first error?

**A** Line **I**

**B** Line **II**

**C** Line **III**

**D** Line **IV**

**E** Line **V**

**F** There is no error, the proof is correct as written.

*Integration*

**13.** For each positive integer $n$, define

$$F(n) = \int_0^1 x^n \, \mathrm{d}x$$

and

$$H(n) = \sum_{r=1}^{n} \frac{1}{F(r)}$$

What is the smallest positive integer $n$ such that $H(n) > 15$?

**A** 3

**B** 4

**C** 5

**D** 6

**E** 7

*Logic*

**14.** Let $S$ be the statement:

$$\text{For all real numbers } x, \text{ if } |x| \geq 1 \text{ then } g(x) > 0$$

Which of the following is a correct **negation** of $S$?

**A** There exists a real number $x$ with $|x| \geq 1$ such that $g(x) \leq 0$

**B** There exists a real number $x$ with $|x| > 1$ such that $g(x) < 0$

**C** For all real numbers $x$, either $|x| < 1$ or $g(x) > 0$

**D** For all real numbers $x$ with $|x| \geq 1$, $g(x) \geq 0$

**E** There exists a real number $x$ such that $|x| < 1$ and $g(x) \leq 0$

**F** There exists a real number $x$ such that $|x| < 1$ or $g(x) \leq 0$

*Proof & Error Analysis*

**15.** A function $f$ is defined for all real $x$, and $f'(x)$ exists for every real $x$.

The function $f$ is also **strictly increasing:** whenever $a < b$, also $f(a) < f(b)$.

A student makes the following claim:

**Claim:** It follows that $f'(x) > 0$ for every real $x$

The student argues as follows:

**Line 1:** Suppose, for a contradiction, that $f'(c) \leq 0$ for some real number $c$

**Line 2:** If $f'(c) < 0$, then just to the right of $c$ the graph of $f$ falls below $f(c)$, which contradicts $f$ being strictly increasing. Hence $f'(c) = 0$

**Line 3:** So the tangent to the graph of $f$ at $x = c$ is horizontal

**Line 4:** A horizontal tangent at $x = c$ means there are numbers $a < c < b$ with $f(a) \geq f(c)$ or with $f(b) \leq f(c)$

**Line 5:** Either possibility contradicts $f$ being strictly increasing. Hence $f'(x) > 0$ for every real $x$

Which of the following best describes the argument?

**A** The argument is correct, and the claim is true

**B** The first error occurs in Line 2

**C** The first error occurs in Line 3

**D** The first error occurs in Line 4

**E** The first error occurs in Line 5

**F** The argument contains an error, but the claim is nevertheless true

*Sequences & Series*

**16.** Let $n$ be an integer with $n \geq 6$.

In the expansion of $(1 + x)^n$ the coefficients of $x^4$, $x^5$ and $x^6$, in that order, are consecutive terms of an arithmetic sequence.

What is the sum of all possible values of $n$?

**A** 13

**B** 14

**C** 20

**D** 21

**E** 22

**F** 28

*Logic*

**17.** Let $a$ be a real number. Consider the following statement:

For all real numbers $x$, there exists a real number $y$ such that
$a \leq y \leq a + \pi$ and $\sin y = \sin x$

Which condition on $a$ is **necessary and sufficient** for the statement to be true?

**A** $\sin a = 0$

**B** $\cos a = 0$

**C** $\sin 2a = 0$

**D** $\cos 2a = 0$

**E** $\sin a = \cos a$

**F** $\sin a = -\cos a$

*Logic*

**18.** Let $p$ and $q$ be real constants, and let

$$f(x) = x^3 + px^2 + qx$$

Consider the statement

S: There exists a real number $x$ such that $f'(x) < 0$

Which of the following conditions is **necessary** but **not sufficient** for S?

**A** $q < 0$

**B** $q \leq 0$

**C** $3q < p^2$

**D** $3q \leq p^2$

**E** $p^2 > 0$

**F** $p^2 \geq 3|q|$

*Algebra*

**19.** A real number $x$ satisfies

$$x^2 - 3x + 1 = 0$$

What is the value of $x^6 + \dfrac{1}{x^6}$?

**A** 288

**B** 306

**C** 320

**D** 322

**E** 324

*Polynomials*

**20.** The equation

$$x^3 - 3x + p = 0$$

where $p$ is a real constant, has three distinct real solutions for $x$.

Find the complete set of values of $p$

**A** $-3 < p < 3$

**B** $-2 < p < 2$

**C** $-2 \leq p \leq 2$

**D** $p < -2$ or $p > 2$

**E** $0 < p < 4$

**F** $-4 < p < 0$